EXPERIMENT - 3

AIM: Create an Employee–Department–Project schema. Insert at least 30 employees across 5 departments and 8 projects. Write SQL queries demonstrating selection, projection, aggregates, GROUP BY, HAVING, CASE expressions, and ORDER BY.

SOURCE CODE:

```
CREATE TABLE Department (
   dept_id INT PRIMARY KEY,
   dept_name VARCHAR(50)
);

CREATE TABLE Project (
   project_id INT PRIMARY KEY,
   project_name VARCHAR(50),
   dept_id INT,
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id)
);

CREATE TABLE Employee (
   emp_id INT PRIMARY KEY,
   emp_name VARCHAR(50),
   dept_id INT,
   project_id INT,
   salary DECIMAL(10,2),
   city VARCHAR(50),
   FOREIGN KEY (dept_id) REFERENCES Department(dept_id),
   FOREIGN KEY (project_id) REFERENCES Project(project_id)
);

INSERT INTO Department VALUES
(1, 'HR'), (2, 'Finance'), (3, 'IT'), (4, 'Marketing'), (5, 'Operations');

INSERT INTO Project VALUES
(101, 'Recruitment System', 1), (102, 'Payroll Automation', 2),
(103, 'Cybersecurity Upgrade', 3), (104, 'Website Revamp', 3),
(105, 'Ad Campaign', 4), (106, 'Market Research', 4),
(107, 'Logistics Optimization', 5), (108, 'Inventory System', 5);

INSERT INTO Employee VALUES
(1, 'Alice', 1, 101, 45000, 'Delhi'),
(2, 'Bob', 1, 101, 42000, 'Mumbai'),
(3, 'Charlie', 1, 101, 47000, 'Pune'),
(4, 'David', 2, 102, 55000, 'Delhi'),
(5, 'Eva', 2, 102, 60000, 'Lucknow'),
(6, 'Frank', 2, 102, 52000, 'Chennai'),
(7, 'Grace', 3, 103, 70000, 'Bangalore'),
(8, 'Hannah', 3, 103, 68000, 'Hyderabad'),
(9, 'Ian', 3, 104, 72000, 'Delhi'),
(10, 'Jack', 3, 104, 65000, 'Pune'),
```

```
(11, 'Karen', 3, 104, 64000, 'Mumbai'),
(12, 'Leo', 4, 105, 48000, 'Delhi'),
(13, 'Mia', 4, 105, 50000, 'Lucknow'),
(14, 'Nina', 4, 106, 51000, 'Chennai'),
(15, 'Oscar', 4, 106, 53000, 'Delhi'),
(16, 'Paul', 5, 107, 56000, 'Hyderabad'),
(17, 'Quinn', 5, 107, 57000, 'Bangalore'),
(18, 'Rita', 5, 107, 59000, 'Delhi'),
(19, 'Sam', 5, 108, 60000, 'Lucknow'),
(20, 'Tina', 5, 108, 61000, 'Mumbai'),
(21, 'Uma', 5, 108, 62000, 'Chennai'),
(22, 'Victor', 2, 102, 53000, 'Delhi'),
(23, 'Wendy', 2, 102, 54000, 'Hyderabad'),
(24, 'Xavier', 3, 103, 71000, 'Bangalore'),
(25, 'Yara', 3, 104, 66000, 'Lucknow'),
(26, 'Zane', 4, 105, 49000, 'Delhi'),
(27, 'Aditi', 4, 106, 52000, 'Mumbai'),
(28, 'Bhavya', 1, 101, 46000, 'Chennai'),
(29, 'Chetan', 1, 101, 43000, 'Lucknow'),
(30, 'Devika', 5, 107, 58000, 'Delhi');

SELECT *
FROM Employee
WHERE city = 'Delhi';

SELECT emp_name, salary
FROM Employee;

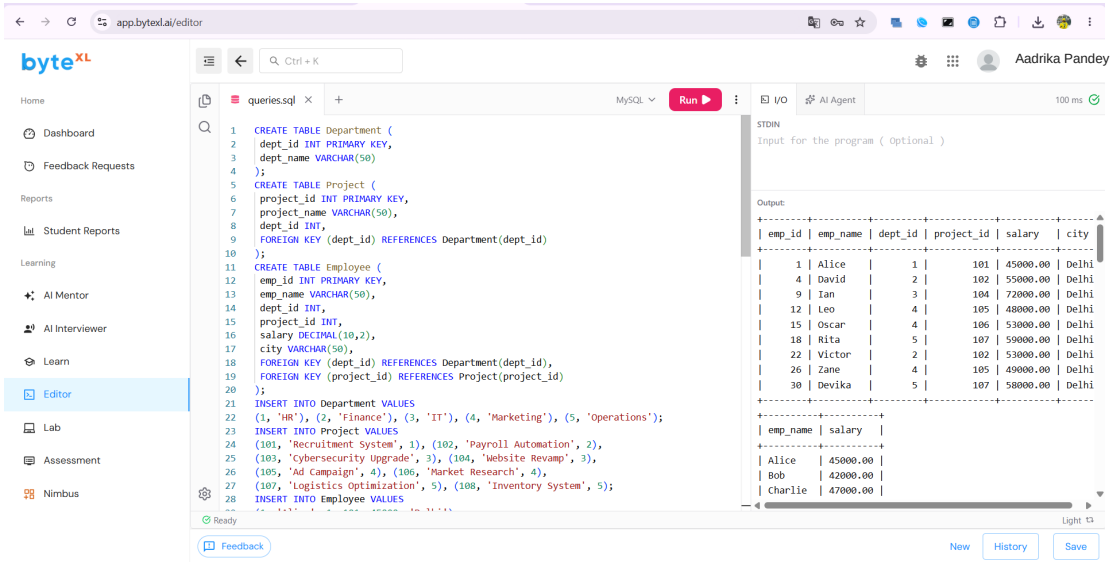
SELECT dept_id, AVG(salary) AS avg_salary
FROM Employee
GROUP BY dept_id;

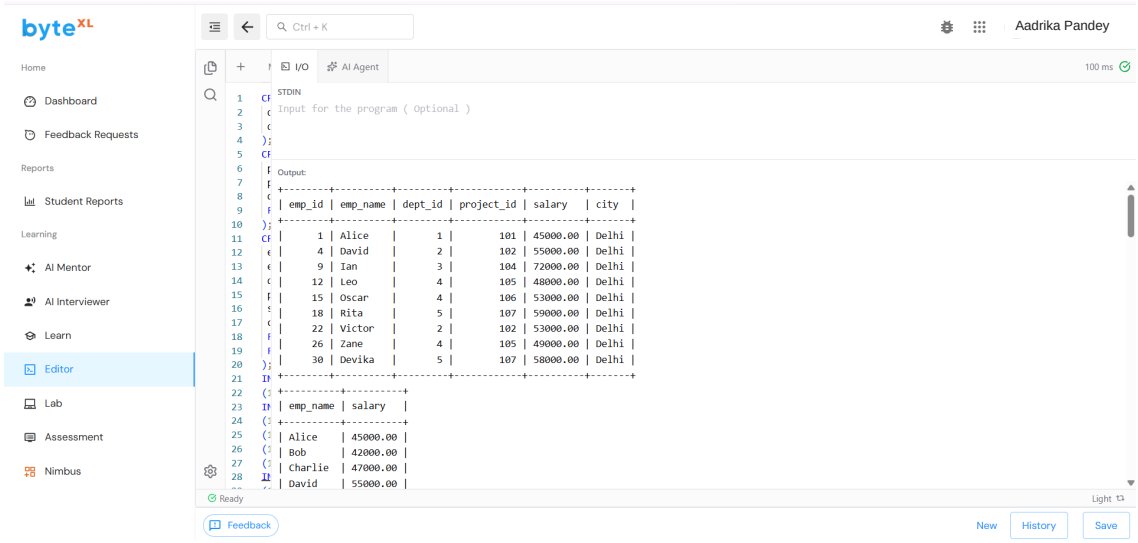
SELECT dept_id, COUNT(*) AS num_employees
FROM Employee
GROUP BY dept_id
HAVING COUNT(*) > 5;

SELECT emp_name, salary,
    CASE
        WHEN salary < 50000 THEN 'Low'
        WHEN salary BETWEEN 50000 AND 60000 THEN 'Medium'
        ELSE 'High'
    END AS salary_category
FROM Employee;

SELECT emp_name, dept_id, salary
FROM Employee
ORDER BY salary DESC;
```

OUTPUT :-


app.bytexl.ai/editor
byteXL
Aadrika Pandey
Ctrl + K
Home
Dashboard
Feedback Requests
Reports
Student Reports
Learning
AI Mentor
AI Interviewer
Learn
Editor
Lab
Assessment
Nimbus
queries.sql
MySQL
Run
I/O
AI Agent
100 ms
STDIN
Input for the program (Optional)
Output:
emp_id	emp_name	dept_id	project_id	salary	city
1	Alice	1	101	45000.00	Delhi
4	David	2	102	55000.00	Delhi
9	Ian	3	104	72000.00	Delhi
12	Leo	4	105	48000.00	Delhi
15	Oscar	4	106	53000.00	Delhi
18	Rita	5	107	59000.00	Delhi
22	Victor	2	102	53000.00	Delhi
26	Zane	4	105	49000.00	Delhi
30	Devika	5	107	58000.00	Delhi
emp_name	salary
Alice	45000.00
Bob	42000.00
Charlie	47000.00
Ready
Light
Feedback
New
History
Save


byteXL
Aadrika Pandey
Ctrl + K
Home
Dashboard
Feedback Requests
Reports
Student Reports
Learning
AI Mentor
AI Interviewer
Learn
Editor
Lab
Assessment
Nimbus
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Ready
Light
Feedback
New
History
Save